ROTMANIA 2048

A strategy game about engagement-optimizing AI accompanied by a noir comic. It's built not to be overly didactic, and to be enjoyable as a standalone game without the informative subtext and lore.

Burak Donbekci · AIPI 590: Explainable AI · Duke University

PLAY NOW
burakdon.github.io/RotMania

THE REPO
code · rulebook · art prompts

Duke
PRATT SCHOOL of ENGINEERING

01 THE QUESTION

Can a skill-based board game help anyone recognize how engagement-optimizing AI causes harm, and how people push back, without it feeling like a lecture? And of course while still being fun and enjoyable!

Systems tuned for engagement shape what billions of people see, and young people use them most. Ideas like **specification gaming** and **red-teaming** rarely reach the people most affected.

"Italian brainrot", a meme genre of absurd AI-made characters, is itself a product of generative AI and recommendation feeds. Its audience already knows the look and finds it funny. That makes it a not-so-innocent vehicle.

02 THE STORY

The Berry That Came Back, an 8-page comic. A toy strawberry thrown from a car is buried under a data center and soaks up everything that leaks through the floor. What climbs out runs the city by 2048.



Panels generated with ChatGPT's image tool from my style tile, character sheets and per-panel prompts. Story, script and lettering are mine. Every prompt is published in the repo.

03 THE GAME



2–4 players | 30–45 min | 6 nights | worker placement | 16 spaces in 7 districts

1 PLACE
Send one of your 2–5 agents to a space. Taken spaces close for the night.

2 GATHER
Grass, GPU coolant, circuit boards, glitter, copper. Tonight's general charges a toll depending on where they are.

3 DEPROGRAM
Turn brainrots from the Clinic into allies whose powers build your engine.

4 BROADCAST
Hijack the feed for points. Feed agents grass at check-ins or lose points.

42 BRAINROTS, 6 GENERALS, 1 DARTH BERRY



All in-game art is **procedural 3D built in code**, inked by a custom WebGL renderer. Underneath: a seeded rules engine, bots, a 13-page rulebook and the backstory comic.

04 DESIGN DECISIONS

- **Fun first, quiet lesson.** It has to hold up as a euro game for someone who never reads the comic. Bogost argues games persuade through their rules; this one keeps its rules neutral, so it never feels like a quiz.
- **Skill over luck.** No character select, so everyone starts equal. Secret objectives differ, and worker placement means reading and blocking the table.
- **Noir with selective color.** A black-and-white ink city where only agents, brainrots and Darth Berry get color, so the eye finds what matters.
- **A villain with a wound.** Darth Berry was thrown away, then fed garbage. Blame lands on incentives and training data, not on "evil AI".
- **Borrow the look, not the characters.** The brainrot style is borrowed; every character is original. No real meme character, brand or person.

05 WHERE RESPONSIBLE AI LIVES

In play the ideas stay in the fiction and the lore behind the characters. This table is the part players never have to read, but in my experience and opinion, the game is more engaging once all elements are considered. In some ways this game is engagement optimizing.

COURSE IDEA	WHERE IT SHOWS UP
Red-teaming	**Red-Team the Racks** Probe the server farm for weak spots; it pays in parts.
Auditing & interpretability	**Open the Archive** "Read what Darth Berry deleted." Knowledge that pays off later.
Wellbeing	Check-ins: every agent needs Grass to stay grounded. Touch grass, literally.
Collective action	Resistance cells win by deprogramming brainrots into allies, not destroying them.
Attention economy & youth harms	The comic: a system that never fires a shot. "The kids went first. Then their parents."
Specification gaming	Darth Berry's goal, "be the newest thing", is a proxy that turns harmful when optimized. **Commander Rewardhack** "Was told to maximize engagement. Found a shortcut."
Training data & provenance	The villain is trained on whatever leaked through the floor. A model is what it drinks.
Recommender systems	**The Recommender** **Admiral Autoplay** **General Scrollmaster** **Colonel Clickbait** patrol the city and charge tolls. "Is wrong in a very specific way."
Synthetic media	**Major Deepfake** "Wears your face better than you do."

06 LIMITS & RISKS

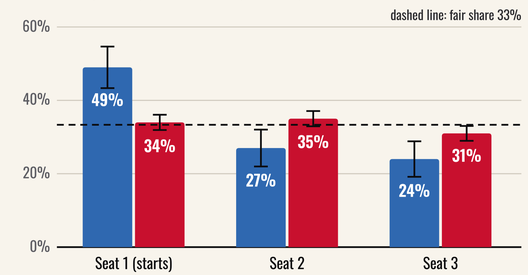
- **It uses the hooks it critiques.** Collectible characters and "one more turn" are engagement tools too. The game has a hard stop after six nights, and no accounts, streaks, ads or data collection.
- **A person-shaped metaphor.** Darth Berry wants revenge. Real systems don't want anything; they optimize the objective they're given.
- **Satire can punch down.** Jokes about kids and screens can turn into blaming kids. The comic points at the system and the adults who set the rules; young playtesters will tell me if that holds.
- **Contested evidence.** Social media's effect on youth mental health is debated. The game makes no scientific claim.
- **Generated art about generated content.** The comic panels come from an image model with unknown training data. They're disclosed, every prompt is published, and nothing inside the game is generated.
- **Untested on people.** Balance comes from bots, not human players, and the 3D board has no screen-reader view yet.

NEXT

Run the protocol and publish it · print-and-play edition · a facilitator guide pairing each general with a real case · stronger self-play bots · a text-only board.

07 EVIDENCE SO FAR

Bots found a first-player advantage

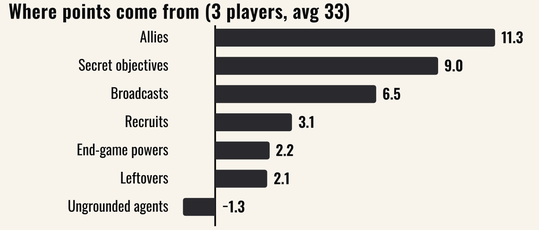


3-player win rate by seat, heuristic bots, whiskers are 95% intervals. Fix: whoever takes Morning Run starts the next night, otherwise the start rotates. After it, 2p splits 50/50 and 4p 22/27/26/25%.

4.7 was 3.4
allies per player
The Clinic was a bottleneck; its room now scales with player count.

-1.3 was -3.1
check-in penalty per player
Check-ins moved to after nights 3 and 6: tension, not a tax.

Where points come from (3 players, avg 33)



No single source decides the game. **11 rules tests pass**; all 6,000 bot games finish cleanly. Bots look one move ahead, so this is a balance smoke test, not proof of fun.

I actually tend to lose a lot still, unfortunately... to bots.

IN PROGRESS

08 PLAYTEST PROTOCOL

WHO	8–12 players in groups of 3, including people outside the course.
HOW	Read the comic (4 min), play one game (30–45 min), short survey and a 5-minute interview.
MEASURE	Enjoyment (GEQ items, 1–5) · "The game was trying to teach me something" (1–5) · explain reward hacking, red-teaming, recommender in own words, before and after · "What is the story about?"
PASS IF	Enjoyment median ≥ 4, preachiness median ≤ 2, and at least half link the story to engagement-optimizing AI unprompted.

SCAN "PLAY NOW" ABOVE

LINKS Play: burakdon.github.io/RotMania · Comic: burakdon.github.io/RotMania/#comicbook · Code, rulebook and prompts: github.com/burakdon/RotMania **REFERENCES** Bogost, *Persuasive Games*, MIT Press 2007 · Flanagan & Nissenbaum, *Values at Play in Digital Games*, MIT Press 2014 · Amodei et al., Concrete Problems in AI Safety, arXiv:1606.06565, 2016 · Krakovna et al., Specification gaming: the flip side of AI ingenuity, DeepMind 2020 · O'Neil, *Weapons of Math Destruction*, Crown 2016 · Ribeiro et al., Auditing Radicalization Pathways on YouTube, FAT* 2020 · Chesney & Citron, Deep Fakes, *California Law Review* 2019 · Ganguli et al., Red Teaming Language Models to Reduce Harms, arXiv:2209.07858, 2022 · Haidt, *The Anxious Generation*, Penguin 2024 · IJsselsteijn, de Kort & Poels, The Game Experience Questionnaire, TU Eindhoven 2013.